Reading the Market: News-Augmented Regime Discovery with Unsupervised Representation Learning

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I. Proposal Updates and Clarifications

LDA topic quality. A two-tier domain stopword list was added and `TABULAR-MATERIAL` articles were excluded to reduce boilerplate contamination. Several topics still contain law-firm language; perplexity improves only marginally from K=10 to K=20 (1,163 → 1,150). K=20 is retained as the primary LDA representation.

BERTopic on daily mean embeddings. Averaging up to 50 FinBERT headline embeddings per day substantially compresses within-day variance, causing HDBSCAN to find only ≈4 natural clusters before the `nr_topics` constraint is applied. BERTopic is retained as a secondary result; the hybrid pipeline uses daily sentiment scalars as the primary text representation.

Hybrid fusion scope. The proposal mentioned structured joint representations as a potential direction. In practice, feature concatenation with joint standardization proved tractable and interpretable for a first pass, and was adopted as the fusion strategy.

Text-only regime stability. Early evaluation revealed that all text-only models produce mean regime durations of 2–9 days, well below any economically meaningful threshold. This was not anticipated in the proposal and motivates the optimization directions in Section II.

II. Progress and Initial Results

2.1 Overall Progress

An end-to-end three-pipeline workflow has been built covering data ingestion, feature engineering, model fitting, and initial evaluation. Market data spans 6,269 trading days (CRSP/FRED, 2000–2024); RavenPack news covers 9,496 calendar days after entity and relevance filtering. Three text representations (sentiment scalars, LDA topics, FinBERT embeddings) and three model classes (GMM, HMM, k-means) were applied across all pipelines, yielding 10 model configurations, each evaluated on four dimensions: statistical fit, clustering quality, temporal stability, and financial interpretability.

2.2 Initial Evaluation Results

The central finding is that **adding news signals does not yet yield a clear improvement over the price-only baseline**. Figure 1 shows that text-only GMM and k-means models switch regimes nearly every day on average, while HMM-constrained models achieve economically plausible multi-week persistence.

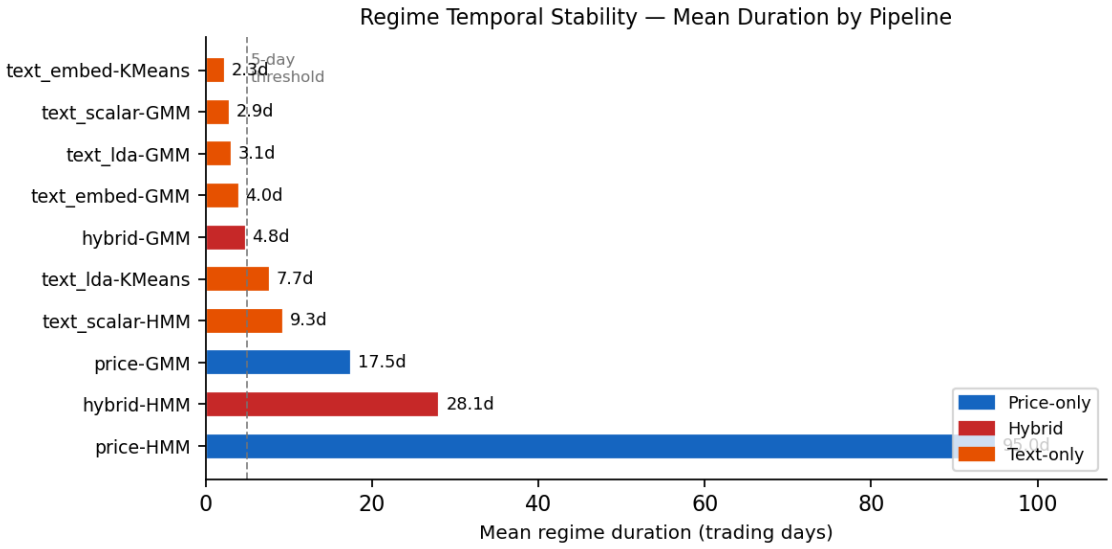


Figure 1: Mean regime duration (trading days) for all 10 pipeline–model configurations.

Table 1 reports the full numerical results across all four evaluation dimensions. Across every metric, price-only models (especially price-HMM) consistently outperform or match the text-augmented alternatives:

- **Temporal stability.** price-HMM leads with a mean duration of 95.0 days and rolling ARI of 0.447. Hybrid-HMM partially recovers stability (28.1 days), but this benefit stems from the HMM temporal prior—hybrid-GMM without it collapses to 4.8 days.
- **Clustering quality.** price-GMM achieves the best silhouette score (0.247). Adding text features dilutes within-regime compactness (hybrid-GMM: 0.118), suggesting the extra dimensions introduce more noise than signal.
- **Financial interpretability.** price-HMM produces the most differentiated regime profiles (Sharpe range: −0.48 to +2.02; mean VIX: 13–41). Text-only regimes of 2–4 days yield economically implausible per-regime statistics.

Type	Pipeline	Silhouette↑	Duration (days)↑	Rolling ARI↑	Best Sharpe↑
Price-only	price-GMM	0.247	17.5	0.369	0.82
	price-HMM	0.202	**95.0**	**0.447**	**2.02**
Text-only	text_scalar-GMM	0.232	2.9	0.433	8.06*
	text_scalar-HMM	0.189	9.3	0.352	18.02*
	text_lda-GMM	0.000	3.1	0.402	0.70
	text_lda-KMeans	**0.289**	7.7	0.019	0.52
	text_embed-GMM	0.075	4.0	0.354	0.75
	text_embed-KMeans	0.172	2.3	0.298	1.07
Hybrid	hybrid-GMM	0.118	4.8	0.366	5.29*
	hybrid-HMM	0.091	28.1	0.291	1.39

Table 1: Four-dimensional evaluation of all 10 model configurations. Bold marks the best value per column (excluding *). * Best-regime Sharpe is inflated by a degenerate micro-regime capturing a small number of atypical days; not economically meaningful.

2.3 Technical Issues

Sentiment scalars are too high-frequency. Daily CSS mean reflects single-day news sentiment, which is noisy and mean-reverting. Without temporal smoothing, the clustering algorithm effectively assigns distinct labels to nearly every day.

LDA expressiveness is limited. Daily document aggregation and remaining boilerplate tokens restrict topics to capturing article-volume composition rather than the macro-narrative shifts that correspond to genuine regime transitions.

FinBERT embedding collapse. Averaging 50 headline embeddings per day concentrates all trading-day points near the corpus centroid (PCA(20) retains ≈82% variance), leaving little discriminative signal for downstream clustering.

2.4 Next Steps

- **Temporal smoothing.** Apply 5- or 21-day rolling windows to sentiment scalars to match signal time scale to regime dynamics.
- **Improved text feature selection.** Filter to macro-relevant article types (central bank, macro indicators, credit events) to improve LDA topic quality.
- **Longer embedding aggregation windows.** Experiment with weekly or bi-weekly FinBERT averages to recover variance.
- **Regime characterization (notebook 08).** Complete historical episode mapping, qualitative regime labeling, and the annual regime-composition chart for the final report.